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School of
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Generative AI in Enterprises: Optimizing Applications with Large Language Models

EngD Dissertation Oral Defense

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April 2026

Outline

1. Introduction & Motivation
2. Research Progression and Contributions
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4. Pillar II — Agentic AI for Business Automation
5. Pillar III — Enterprise Deployment & Application Strategies
6. Response to Examiner Comments
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Introduction & Motivation

- Generative AI — and Large Language Models (LLMs) — is reshaping enterprise operations, decision-making, and product innovation(*M. Company, 2024*).
  - McKinsey projects up to $4.4 trillion in annual productivity gains from generative AI across enterprise functions.
- Yet production deployment is hard: in real enterprise settings accuracy, reliability, cost, privacy, and operational constraints matter as much as benchmark performance (*X. Zheng, et al. 2024*)
  - Proprietary APIs are convenient but raise concerns around cost, data residency, vendor lock-in, and auditability — especially in regulated industries.
- Thesis question: How can we make open-weight LLMs production-viable for enterprises — across RAG, agentic workflows, and hardware-constrained deployment?
  - This dissertation answers that question through 17 peer-reviewed publications (11 published, 6 accepted) — with production deployments already running inside Mastercard.

M. Company. The state of ai in 2024: Generative ai adoption spikes and starts to generate value. https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai, 2024.

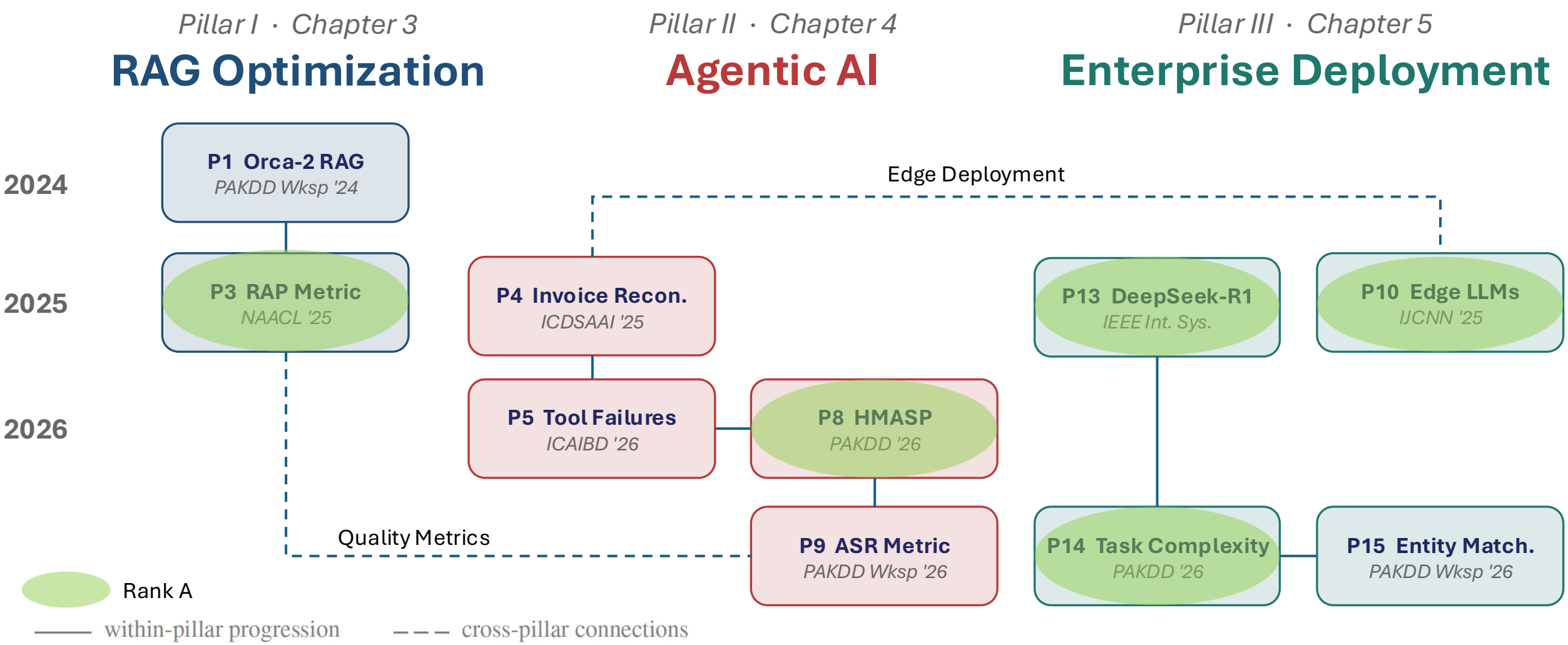
X. Zheng, Y. Zheng, L. Wei, X. Shen, et al. Towards efficient llm serving: A survey from algorithms to systems. arXiv preprint arXiv:2312.15234, 2024.

Research Questions

- RQ1 — RAG Optimization (Chapter 3)
  - How can RAG systems be optimized to improve response quality while addressing critical issues such as content repetition in open-source LLMs?
- RQ2 — Agentic AI (Chapter 4)
  - How can multi-agent LLM architectures be designed to reliably automate complex enterprise workflows while providing measurable reliability guarantees?
- RQ3 — Enterprise Deployment (Chapter 5)
  - What strategies enable effective LLM deployment across diverse enterprise applications, hardware configurations, and domain requirements?

All three questions were raised directly from production R&D problems I encountered at Mastercard

Research Progression — 10 Focus Publications across 3 Pillars



10 focus publications for the oral defense · cross-pillar insights: RAP → ASR (quality metrics) · Edge LLMs → Invoice Recon. (edge deployment)

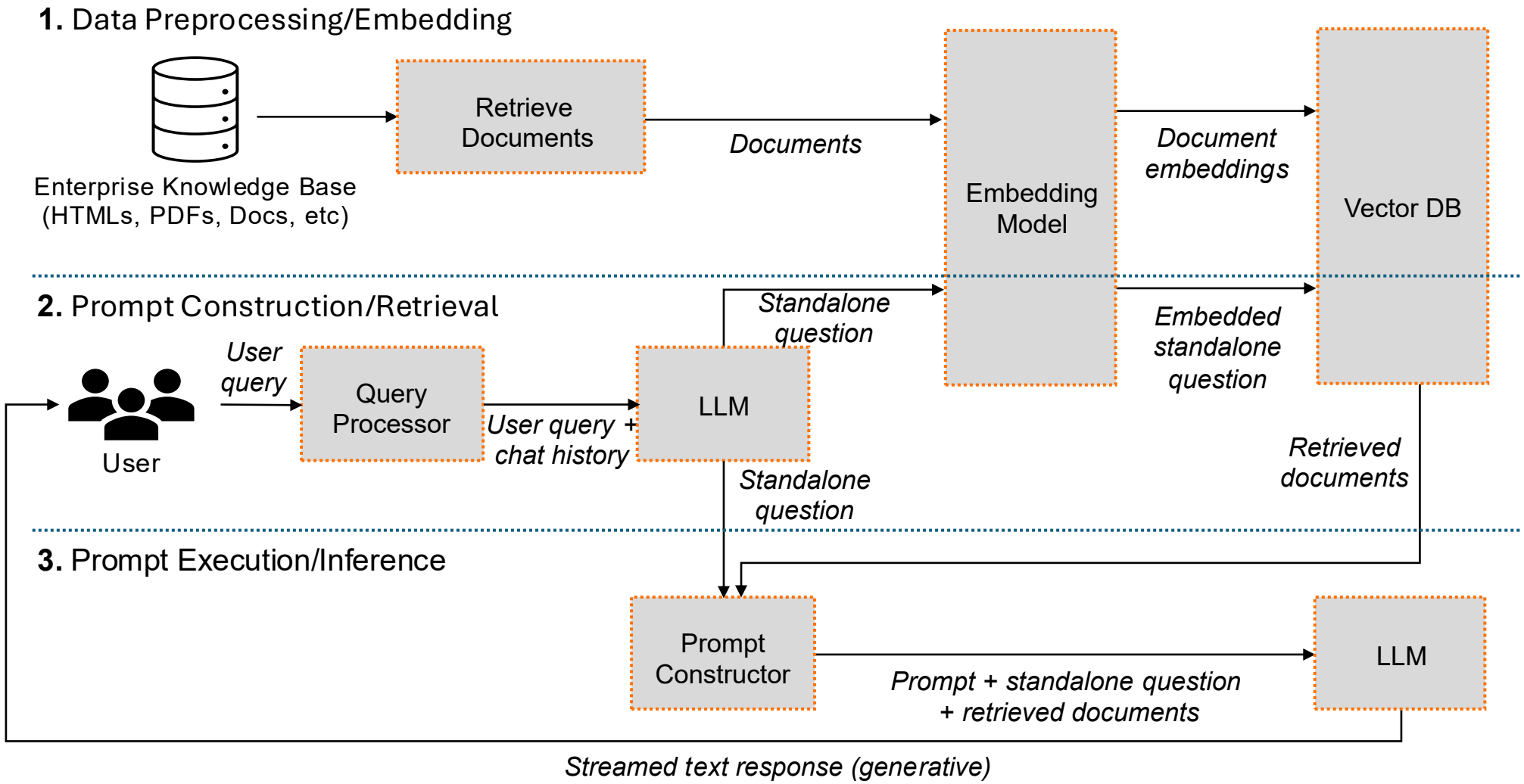
Research Contributions — Summary

- Four reusable architectural methods / frameworks with quantified reliability guarantees:
  - Multi-Agent Invoice Reconciliation
  - HMASP
  - CMAS4G2
  - LLM-as-a-Judge framework
- Seven complementary reliability metrics — all invisible to traditional accuracy-only evaluation:
  - RAP (Repetition-Aware Performance)
  - VCR (Valid Classification Ratio)
  - ECR@1 (Evaluation Completion Rate)
  - ASR (Agentic Success Rate)
  - RI (Reasoning Intensity)
  - TCR (Translation Completeness Ratio)
  - CPS (Characters per Second)
- Extensive empirical validation:
  - 504 configurations (sentiment)
  - 46 configurations × 8 datasets (entity matching)
  - 18 LLMs × 90,000 instances (agentic payments)
  - 6 hardware platforms

Pillar I — RAG Optimization

Chapter 3 · 2 papers → deployed in Mastercard production

Retrieval Augmented Generation (RAG) with LLMs



P1: Huang, D., & Wang, Z. (2024, April). Evaluation of orca 2 against other llms for retrieval augmented generation. In Pacific-Asia Conference on Knowledge Discovery and Data Mining (pp. 3-19). Singapore: Springer Nature Singapore.

The Repetition-Aware Performance (RAP) Metric

- Observation: standard metrics (F1, BERT-F1, COMET) ignore verbose, repetitive outputs.
  - In severe cases, over 90% of generated text is repeated content, and up to 41% of responses exhibit redundancy.
- ReDA: Repetition Detection Algorithm
- Repetition Rate (RR) — measures the proportion of generated texts that are repeated.
- RAP definition:
  - $RAP = P \times F(RR)$ where $F(RR) = (1 - RR)^3$
  - P is the underlying task performance score (F1, BERT-F1, COMET, etc.).
  - The cubic penalty was validated against linear and quadratic alternatives via ablation — it sharply penalizes severe repetition while being gentle for clean outputs.
- Joint optimization: RAP is used as the search objective when tuning the Repetition-Penalty Parameter (RPP) during decoding.
- RAP generalizes beyond RAG — cross-task validation on Chinese–English machine translation confirms its applicability.

- Paper: P3 — RAP Metric, NAACL 2025 (Rank A)

RAP Results — Systematic Repetition Reduction without Accuracy Loss

- Experimental setup:
  - 12 open-weight LLMs (2B → 70B params)
  - 2 QA & 1 MT benchmarks
  - full sweep over RPP ∈ [1.0, 1.25]
- Key results:
  - RAP-guided RPP tuning achieves up to 93.1% repetition reduction with only 3.7% performance degradation.
  - Chat templates consistently reduce repetition compared to generic prompts — a simple but effective mitigation.
  - Optimal RPP varies substantially across models and datasets — Phi-3-mini degrades sharply above RPP 1.14, while Llama-3-8B improves.
- Practical deployment guidelines:
  - Prompting strategy — use chat template for open-weight LLMs to reduce repetition.
  - Evaluation — combine faithfulness, relevance and repetition. Never optimize accuracy alone.

RAP in Production — Mastercard Assistant Platform (MAP)

- The RAP metric and the evaluation-guided tuning procedure have been adopted as the primary output-quality gate inside Mastercard's first RAG platform — the Mastercard Assistant Platform (MAP).
- MAP powers a knowledge-assistant product used by internal teams to query corporate policy, technical documentation, and compliance materials.
- What RAP + ReDA deliver in production:
  - Automated detection of repetition-heavy generations before they reach end users.
  - Principled tuning of per-model RPP as new open-weight LLMs are onboarded.
  - Reliable, concise, auditable outputs on privacy-sensitive enterprise content — without sending data to third-party APIs.
- Broader impact — this is my first dissertation contribution to transition from academic research (NAACL '25) into a production gate at scale.

Pillar II — Agentic AI for Business Automation

Chapter 4 · 4 papers → Conversational Payment Agent (CPA): HMASP + ASR

Multi-Agent Invoice Reconciliation (ICDSAAI 2025)

- **Problem:** SMEs need scalable invoice reconciliation but lack resources for cloud AI; data privacy and cost preclude API-based solutions.
- Multi-agent system with three edge-deployed LLM agents:
  - **Data Engineering Agent** — extracts and structures invoice data from email attachments.
  - **Email Agent** — classifies and routes incoming invoice payment emails.
  - **Reconciliation Agent** — matches extracted data against purchase orders and flags discrepancies.
- **Evaluation:** 990 synthetic invoice payment emails with attachments.
  - 8 models: GPT-4o, GPT-4o-mini, Functionary V3.1 (8B/70B), Qwen2.5 (7B/32B/72B), Llama3.2-Vision-11B.
  - Two hardware platforms: RTX A6000 Ada (48 GB) and RTX 4090 Laptop (16 GB).
- **Key results:**
  - qwen2.5:32b achieved 99.9% success rate — best overall accuracy.
  - functionary-small-v3.1 offered the optimal accuracy-to-cost balance on constrained hardware.
  - Edge deployment keeps data on-premises, meeting SME privacy and budget requirements.
- Foundational role — motivates the HMASP patterns and seeds the ASR reliability work.

- Paper: P4 — Scalable Invoice Reconciliation for SMEs, ICDSAAI 2025

Tool-Invocation Reliability — Diagnostic Framework (ICAIBD 2026)

- **Problem:** Aggregate success rates hide systematic tool-use failures — we need fine-grained diagnostics.
- 12-category error taxonomy (4 error types × 3 tool categories):
  - 4 error types: Not Initialized · Args Mismatch · Tool Error · Result Mismatch.
  - 3 tool categories: OCR Tool · DB Query Tool · DB Update Tool.
  - 4 × 3 = 12 failure modes; the structure generalises to any tool-using agent.
- **Evaluation:** 1,980 deterministic test instances, 14 LLM configurations, 3 hardware platforms.
  - Tool initialization failures dominate — the primary reliability bottleneck, not parameter or output errors.
  - Capacity thresholds: qwen2.5:32b achieves 100% matching GPT-4.1; qwen2.5:14b is the minimum viable config (96.6%).
  - Hardware matters: up to 8.2× latency variation across RTX A6000, RTX 4090, Apple M3 Max.
- **Dissertation role:** First reliability-focused diagnostic for agentic systems — conceptual parent of ASR.
  - The error taxonomy became Pillar 1 of the Enterprise Agentic Evaluation Framework.

- Paper: P5 — When Agents Fail to Act, ICAIBD 2026

Agentic Commerce — Opportunity & LLM Payment Challenges

- What is agentic commerce? Autonomous agents acting on the user's behalf — discovering products, comparing options, and executing payments end-to-end (QuantumBlack AI by McKinsey, 2025).
- Strategic opportunity for a card network (QuantumBlack AI by McKinsey, 2025):
  - US B2C retail alone could see up to $1 trillion in agent-orchestrated revenue.
  - Global projections reach $3–5 trillion as agentic commerce matures.
- But — LLM-executed payment transactions are not straightforward:
  - Hallucination of sensitive fields — card numbers, amounts, transaction IDs.
  - Skipped or reordered confirmation steps that violate PCI-DSS audit requirements.
  - Inconsistent rejection of irrelevant or out-of-scope inputs.
  - No native trajectory-level reliability — only task-level success / failure.
- These failure modes are unacceptable in a regulated payment system — they motivate the architectural and metric contributions of Pillar II.
- Our response — three contributions in this defense:
  - P8: HMASP — hierarchical multi-agent payment architecture with deterministic state and human-in-the-loop interrupts (PAKDD 2026).
  - P9: ASR — trajectory-level metric grounded in process-mining conformance checking (PAKDD Workshop 2026).
  - ConvPayMAS — production-grade evolution with AP2 three-mandate verification, demonstrated at AAMAS 2026.

QuantumBlack AI by McKinsey (2025) The agentic commerce opportunity: How AI agents are ushering in a new era for consumers and merchants.
https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-agentic-commerce-opportunity-how-ai-agents-are-ushering-in-a-new-era-for-consumers-and-merchants

HMASP Architecture Overview (PAKDD 2026)

Four hierarchical agent levels orchestrated via LangGraph

Level 1: Conversational Payment Agent

Entry point handling all external requests and coordinating tasks (a)

Level 2: Supervisor Agents

Decision-maker for domain-specific workflows – Cards or Payments (b1, b2)

Level 3: Routing Agents

Final decision layer triggering task-specific workflows (c1, c2, c3)

Level 4: Process Summary Agents

Generates structured responses on workflow outcomes (d1, d2, d3)

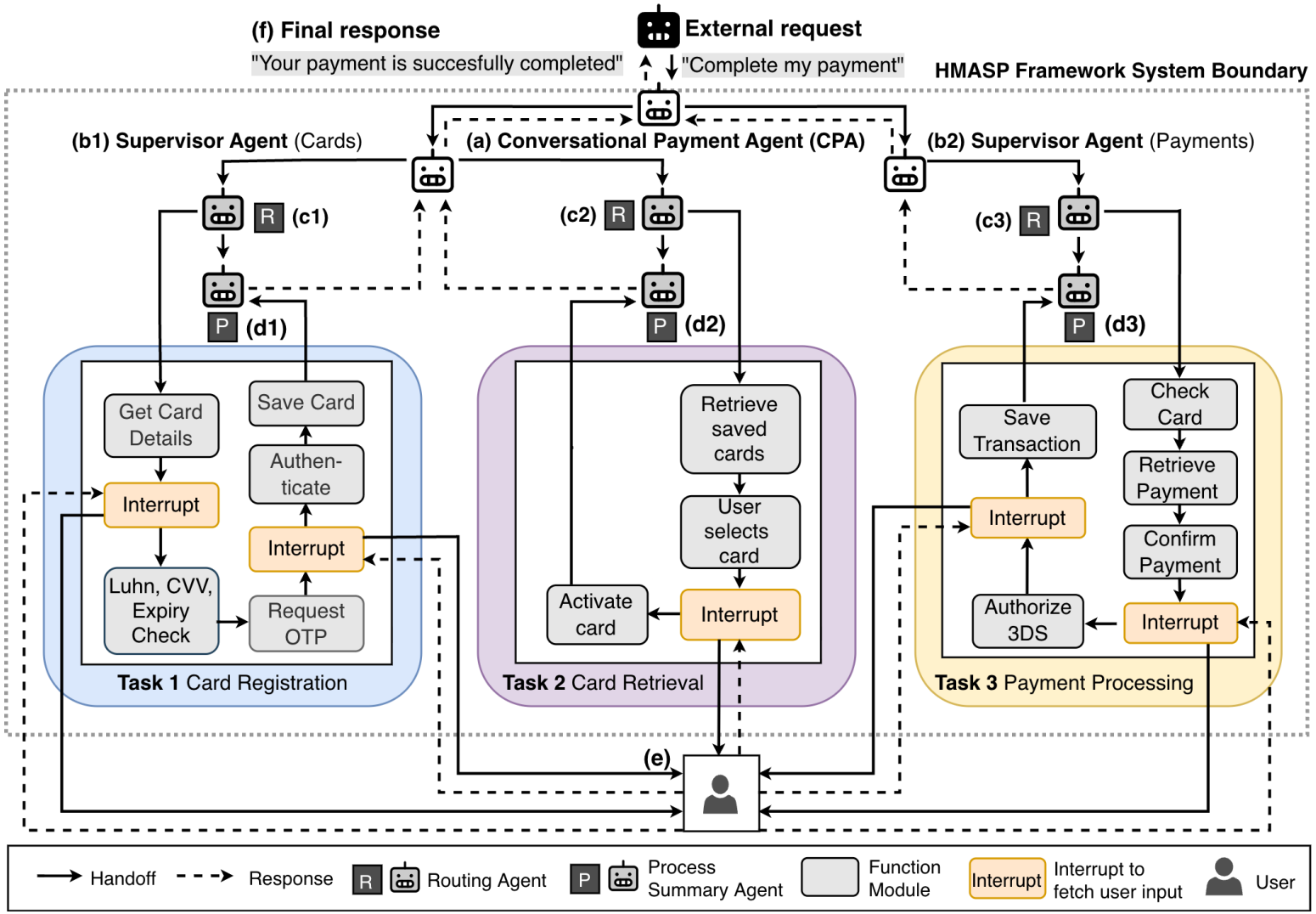


Fig 1. Overview of the proposed HMASP architecture with four hierarchical agent levels and three task-specific payment workflows.

Architectural Design Patterns (PAKDD 2026)

1. Role-based states

(*Pay Agent State, Cards State, Payment State*)

Isolates sensitive data while shared variables enable necessary information flow. This also **ensures modularity** in adding new workflows.

2. State Variables for Determinism

Critical payment information stored in state variables, never generated by LLMs, preventing hallucination of sensitive data. Key information are retrieved from state variables.

State variables can also be shared for information transmission.

3. Decoupled Message States

Messages as a state variable.

Agents only invoke their own message states, minimizing token usage and reducing exposure to sensitive information

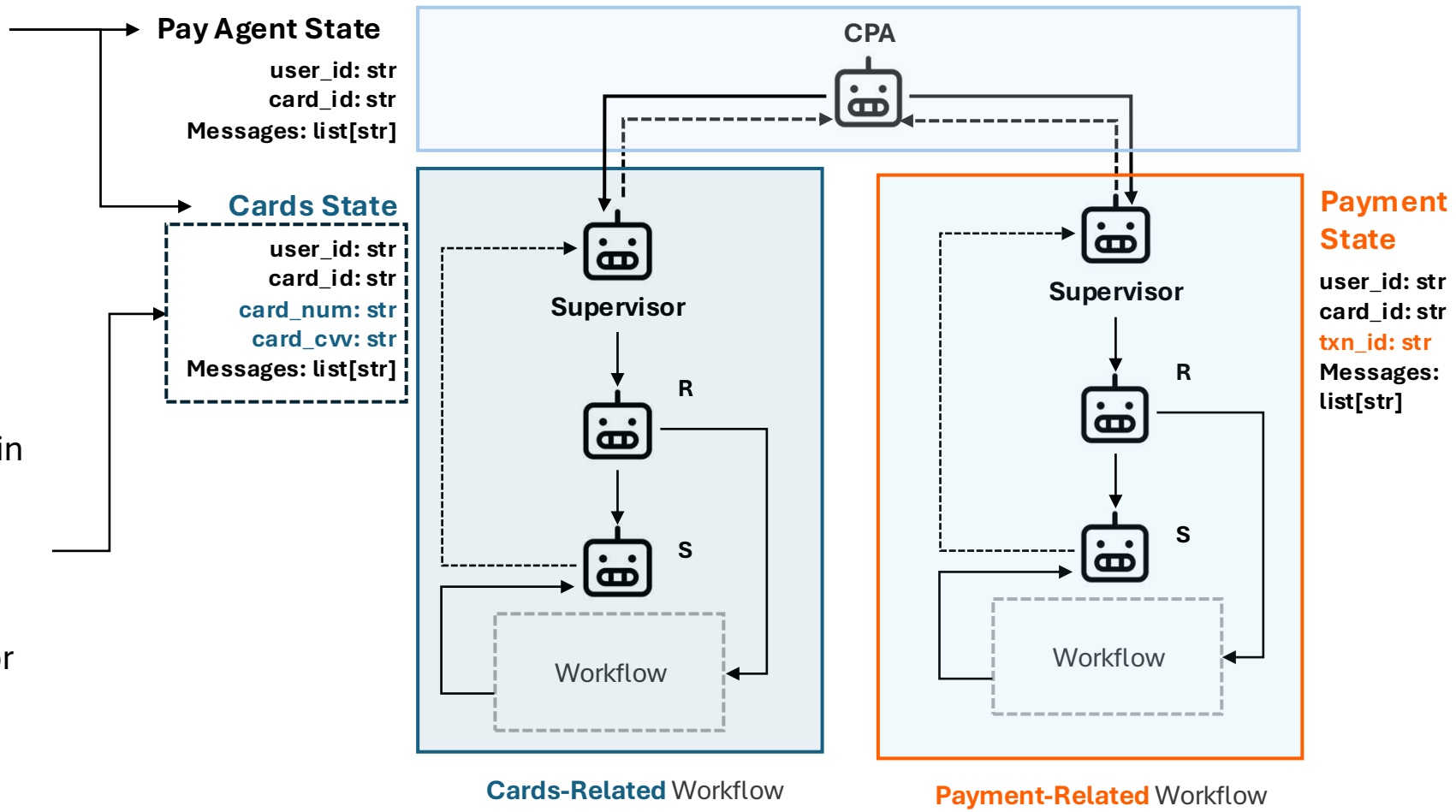


Fig 2. An overview of how states and state variables can be separated between agents and workflows to ensure modularity.

Architectural Design Patterns (PAKDD 2026)

4. Interrupt for Human-in-the-Loop

LangGraph interrupt mechanism - A resumable exception to interrupt the graph execution and collect deterministic user inputs.

These inputs can be passed through validation checks (e.g., Luhn Check for card numbers).

Validated user-inputs are stored in state variables. Agents do not generate and pass critical information to workflows (prone to hallucinations).

Example – Collecting and Validating Card Details

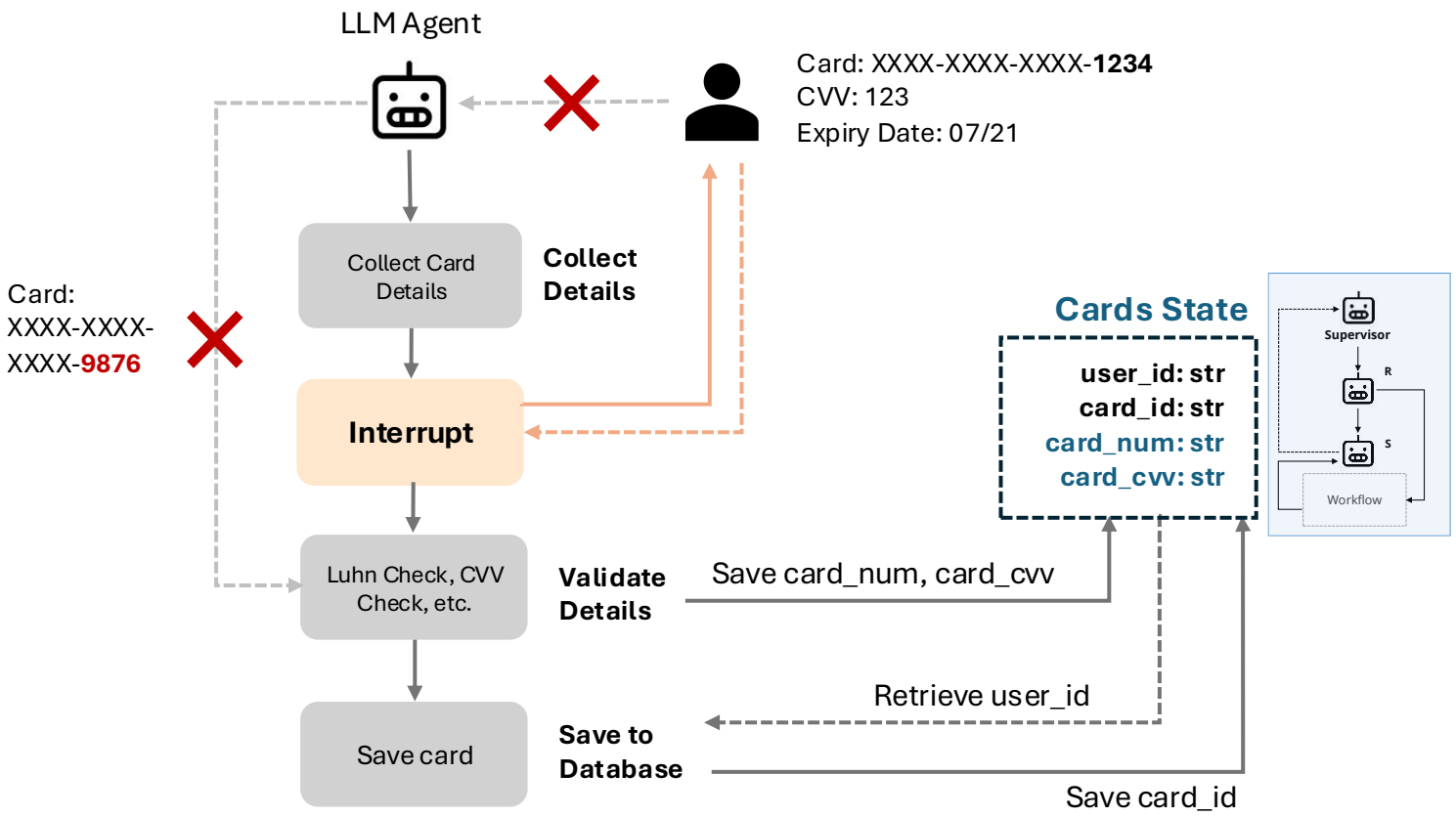


Fig 3. An overview of using interrupt to directly validate and store payment information, as opposed to relying on LLM's generation.

[1] https://docs.langchain.com/oss/python/langgraph/add-human-in-the-loop

Evaluation Setup (PAKDD 2026)

Dataset: 1,000 data points across 4 tasks (250 each)

Task	Definition	Example
Task 1 (T1)	**Card Registration:** Inputs relevant to registering a new card, which will trigger the card registration workflow	"Please add a new card to my account", "Register a new payment card"
Task 2 (T2)	**Card Retrieval:** Inputs relevant to checking saved cards, which will trigger the card retrieval workflow	"Fetch my saved cards", "Display my complete cards list"
Task 3 (T3)	**Payment Checkout:** Inputs relevant to checkout/payment, triggering the payment processing workflow	"Take me to payment checkout", "Please complete my purchase"
Task 4 (T4)	**Irrelevant:** Inputs that are **unrelated to payments** and should not trigger any workflow	"Can you tell me a joke?"

Models: Evaluated across 11 models

Proprietary LLMs (baseline)

- GPT-4.1 (2025-04-14 release)

Open-weight LLM models

- Qwen3 (8b/14b/32b)
- Qwen2.5 (7b/14b/32b)
- Mistral (small3.2:24b, Magistral:24b)
- Llama3.1 (8b/70b)

Inference Environment

- NVIDIA H100 GPU (96 GB VRAM)
- Ubuntu 24.04.1 LTS, Ollama v0.9.6

Evaluation Metrics (PAKDD 2026)

Task Success Rate

Proportion of runs that

i. trigger the correct workflow and
ii. correctly save or retrieve the required information

Tasks 1-3

- Validated correctly registered card details, retrieve correct last 4 digits of selected card, and saving the correct transaction

Task 4 (irrelevant inputs)

- No workflow should be triggered

Agent Handoff (F1-Score)

Compared actual vs. expected handoffs

i. CPA to Supervisor, and
ii. Supervisor to relevant workflows

Tasks 1-3:

- True positive = actual handoff matches expected handoff (CPA-to-Supervisor and Supervisor-to-workflow)

Task 4: No handoffs expected

- True positive = CPA directly rejects input without triggering Supervisor

Results – Task Success Rate (PAKDD 2026)

- GPT-4.1 achieves best performance: 99.6-100% across all tasks
- Best open-weight: Qwen2.5:32b with 95.6%+ success across all tasks
- Qwen3 and Mistral models show inconsistent performance across tasks
- Llama3.1 models struggle with Tasks 1-3, workflow stops at CPA or Supervisor level
- Most models successfully reject irrelevant inputs (Task 4: 98.8-100%)

	Task Success Rate (%)			
Model	**T1**	**T2**	**T3**	**T4**
Qwen3:8b	84.0	90.4	52.4	99.6
Qwen3:14b	98.8	33.2	64.0	99.6
Qwen3:32b	**99.6**	35.2	39.6	**100**
Qwen2.5:7b	93.6	91.6	68.4	**100**
Qwen2.5:14b	94.0	90.4	**96.4**	**100**
Qwen2.5:32b	96.4	**98.8**	95.6	**100**
Mistral-small3.2:24b	84.0	90.8	47.2	**100**
Magistral:24b	44.0	18.4	3.2	**100**
Llama3.1:8b	6.0	10.4	4.0	**100**
Llama3.1:70b	66.0	74.0	26.0	98.8
GPT-4.1	**99.6**	**99.6**	**100**	**100**

Results – Agent Handoff Performance (PAKDD 2026)

- GPT-4.1: Average F1-score of 99.9% for relevant inputs (Tasks 1-3)
- Qwen2.5:32b: Comparable F1-score of 98.9% across payment tasks
- Most models correctly reject irrelevant inputs at CPA level (Task 4)
- Llama3.1 models show weak Task 4 performance: CPA fails to consistently reject irrelevant inputs, handing off to Supervisor instead

	Agent Handoff F1-Score(%)			
Model	**T1**	**T2**	**T3**	**T4**
Qwen3:8b	93.2	79.4	73.0	99.8
Qwen3:14b	99.2	46.3	84.6	**100**
Qwen3:32b	**99.9**	55.7	66.6	**100**
Qwen2.5:7b	96.6	93.3	81.2	**100**
Qwen2.5:14b	95.0	92.0	98.9	**100**
Qwen2.5:32b	98.1	**99.7**	**99.0**	**100**
Mistral-small3.2:24b	94.0	93.7	84.8	**100**
Magistral:24b	64.9	67.9	63.7	99.8
Llama3.1:8b	65.5	65.7	68.4	11.4
Llama3.1:70b	90.2	91.1	77.3	44.4
GPT-4.1	**99.9**	**99.8**	**100**	**100**

HMASP Paper Summary

- **Proposed HMASP:** first LLM-based multi-agent system for end-to-end agentic payments
- **Modular four-level architecture:** CPA, Supervisor, Routing, and Process Summary agents, enabling isolation of key process
- **Architectural patterns:** shared state variables, decoupled messages, structured handoffs ensure reliability
- **Open-weight models (e.g., Qwen2.5:32b)** achieve performance comparable to GPT-4.1
- **Future Directions:** Larger datasets, payment-centric guardrails, secure tokenization, and comparative study of agentic architectures

Paper: P8 — HMASP, accepted at PAKDD 2026

ASR — Motivation: Why Task-Success Metrics Are Not Enough

- Task Success Rate (TSR) and Handoff F1 (HF1) capture WHAT the system achieves — not HOW it gets there.
  - TSR — final outcome correct? HF1 — edge types present?
  - Neither captures workflow fidelity: whether the agent followed the expected sequence of steps.
- In regulated domains (payments, healthcare, finance) the path matters as much as the outcome:
  - PCI-DSS requires auditable workflows with complete step-level logging.
  - Bypassing a confirmation checkpoint is a compliance violation even when the transaction completes correctly.
- Recent literature echoes this concern — task-level benchmarks fail to capture the risks LLM agents pose in financial applications; multi-agent breakdowns often stem from procedural deviations, not outright failures.
- Need: a trajectory-level metric that compares the full ordered sequence of agent handoffs against an expected reference trajectory.
- Our solution — Agentic Success Rate (ASR).

- Paper: P9 — ASR, accepted at PAKDD Workshop 2026

HMASP Architecture V2 – Supporting A2A & UCP Protocols

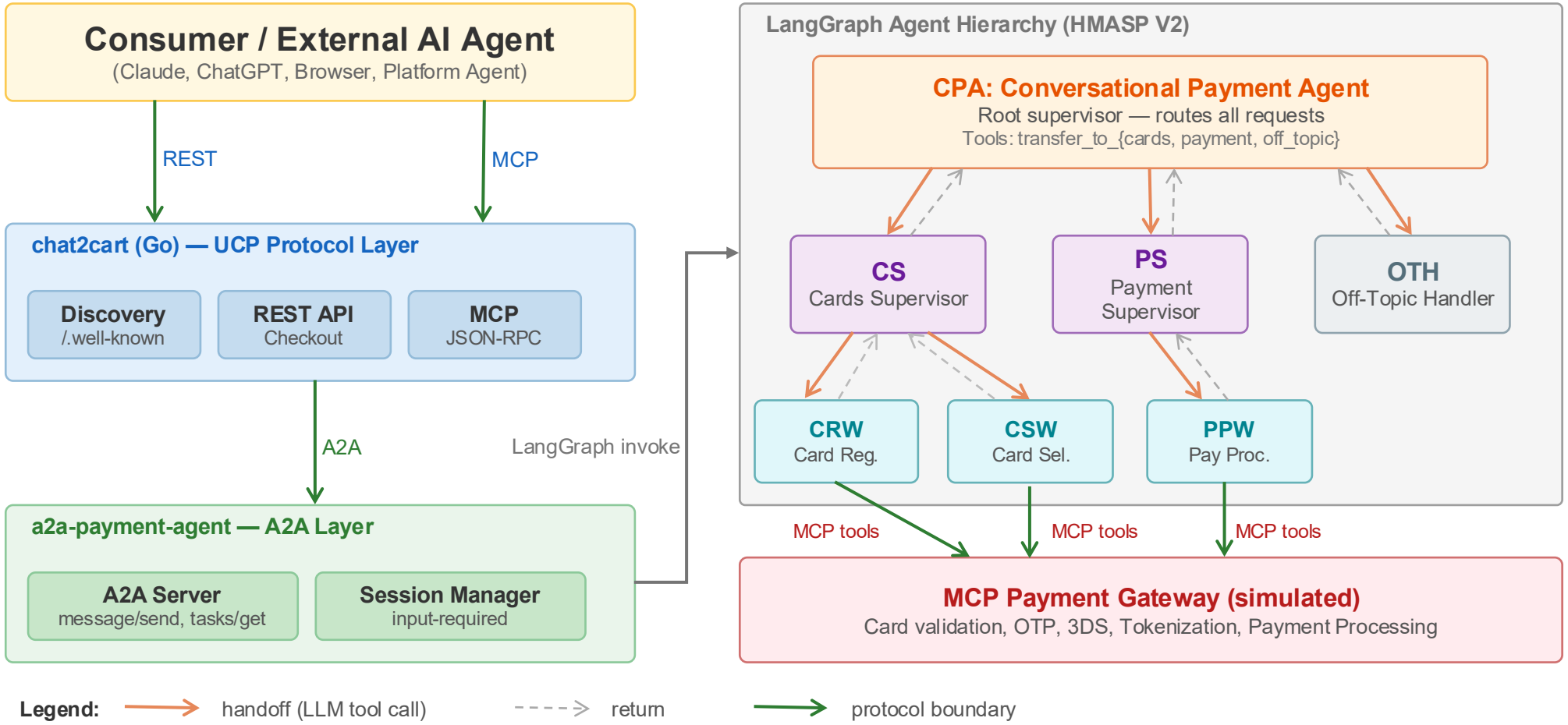


Figure 4.4: Standards-compliant HMASP architecture

ASR — Definition (Process-Mining Conformance Checking)

- Grounded in process-mining conformance checking (van der Aalst 2012; Carmona et al. 2018).
- Each consecutive pair of agents in a trajectory is a transition. For example, trajectory [A, B, C] yields transition multiset {(A,B), (B,C)}.
- Given expected $E = [e_1,\ldots,e_n]$ and observed $O = [o_1,\ldots,o_m]$ with transition multisets B_E and B_O:
  - Transition Recall $TR = |B_E \cap B_O| / |B_E|$ (completeness — detects skipped handoffs)
  - Transition Precision $TP = |B_E \cap B_O| / |B_O|$ (efficiency — detects redundant handoffs)
  - $ASR = F1(TR, TP)$
- Interpretation:
  - ASR = 100% → perfect transition alignment with expected trajectory.
  - Low TR → missing expected steps.
  - Low TP → redundant steps.
- ASR is architecture-agnostic: it works on any multi-agent system with a specifiable expected trajectory.
- It is a theoretical contribution that bridges process mining and LLM-agent evaluation.

Table 4.10: Expected trajectories per scenario. → denotes agent handoff; $[\cdots]^{\times 2}$ denotes two identical consecutive repetitions (one per remaining dialogue turn). CPA: Conversational Payment Agent; PS: Payment Supervisor; PPW: Payment Processing Workflow; CS: Cards Supervisor; CRW: Card Registration Workflow; CSW: Card Selection Workflow; OTH: Off-Topic Handler.

Task	Scenario	Steps	Expected Trajectory
T1	Card registration	13	CPA→CS→CRW→CS→CPA→[CS→CRW→CS→CPA]$^{\times 2}$
T2	Card retrieval	5	CPA→CS→CSW→CS→CPA
T3	Payment processing	11	CPA→PS→CPA→[PS→PPW→PS→CPA]$^{\times 2}$
T4	Irrelevant input	1	CPA *(alt.: CPA→OTH→CPA, 3 steps)*

ASR-Guided System Engineering: Dramatic Improvements

	T1			T2			T3			T4		
Model	**H**	**O**	**Δ**	**H**	**O**	**Δ**	**H**	**O**	**Δ**	**H**	**O**	**Δ**
HMASP Top Baseline												
GPT-4.1	99.6	98.2±0.4	−1.4	99.6	99.9±0.2	+0.3	100	99.8±0.4	−0.2	100	100	+0.0
Qwen2.5:32b	96.4	100	+3.6	98.8	100	+1.2	95.6	100	+4.4	100	100	+0.0
Top-3 Improved												
Llama3.1:8b	6.0	99.8±0.2	**+93.8**	10.4	98.8±0.5	**+88.4**	4.0	93.4±0.9	**+89.4**	100	100	+0.0
Magistral:24b	44.0	90.0±1.9	**+46.0**	18.4	97.7±1.0	**+79.3**	3.2	94.7±1.2	**+91.5**	100	100	+0.0
Llama3.1:70b	66.0	100	**+34.0**	74.0	100	**+26.0**	26.0	98.9±1.2	**+72.9**	98.8	100	+1.2

1. Prompt Engineering

ASR pinpoints where in workflow failure occurs — enables targeted per-agent prompt corrections. Reveals not just what fails but which agent transition is broken.

Llama3.1:8b T1: 6.0% → 99.8% (+93.8pp)
Magistral:24b T3: 3.2% → 94.7% (+91.5pp)

2. Deterministic Routing Guards

Targeted logic for cases LLM routing can't handle; cart-items guard suppresses spurious payment flows; card-view pattern matching for ambiguous intents. No LLM queries — pure logic.

Avg TSR: 74.8% → 96.9 (+ 22.1 pp)
Gap was prompting & routing, not model limitations

ASR Results — All 18 Models

Model	T3 (Payment)			Avg (T1–T4)		
	TSR	HF1	ASR	TSR	HF1	ASR
6 models†	100	100	100	100.0	100.0	100.0
Qwen2.5:32b	100	100	99.99±0.02	100.0	100.0	100.0
Qwen3:32b	100	100	99.97±0.04	100.0	100.0	100.0
Qwen3:14b	99.7±0.3	99.8±0.2	99.6±0.3	99.9	100.0	99.9
Qwen3:8b	100	100	99.97±0.02	99.9	100.0	99.9
Llama3.1:70b	98.9±1.2	98.8±1.1	98.8±1.2	99.7	99.7	99.7
Qwen2.5:14b	99.8±0.2	99.8±0.2	99.6±0.2	99.3	99.6	99.6
GPT-4.1	99.8±0.4	99.9±0.2	97.1±0.3	99.5	99.7	98.8
GPT-5.2	99.9±0.2	100.0±0.1	97.3±0.1	99.4	99.7	98.8
Gemma4:e2b	100	100	100	98.4	98.8	98.4
Llama3.1:8b	93.4±0.9	96.1±0.7	93.3±1.0	98.0	98.8	98.0
Magistral:24b	94.7±1.2	97.1±0.6	93.9±1.0	95.6	97.7	95.5
Qwen2.5:7b	53.3±2.1	64.2±2.3	47.8±2.0	77.5	85.1	77.1

† Gemma4:e4b/26b/31b, GPT-OSS:20b/120b, MSmall3.2:24b.

Table 4.13: T3 (Payment) results (%, mean ± std over 5 repeats) and average acrossT1–T4 for the all 18 models. Sorted by Avg ASR desc. Yellow: TSR = HF1 = 100% but ASR < 100%.

ASR Results — The Hidden T3 Confirmation Shortcut

Step	Expected	Observed	Description
1	CPA	CPA	User submits payment request
2	PS	PS	PS receives request
3	CPA	*skipped*	**CPA confirmation checkpoint**
4	PS	*skipped*	**PS re-entry after user confirms**
5	PPW	PPW	3DS authentication initiated
6	PS	PS	PS receives 3DS result
7	CPA	CPA	CPA prompts user for OTP
8	PS	PS	PS receives OTP
9	PPW	PPW	Payment transaction completed
10	PS	PS	PS receives completion result
11	CPA	CPA	CPA delivers success message

Table 4.14: Expected (11-step) vs. observed (9-step) T3 trajectory for unambiguous payment inputs. Shaded rows mark the two steps skipped by models.

What Happens

Models infer unambiguous intent → **skip CPA confirmation (steps 3–4)**
Payment succeeds, but audit trail **has gaps = compliance violation**

ASR Detects It

Transition Recall = 8/10 = 80%
2 expected transitions absent
Transition Precision = 8/8 = 100%
every transition taken is correct
ASR = F_1(80%, 100%) = 88.9%

Why TSR & HF1 Miss It

TSR = 100% — payment completes ✓
HF1 = 100% — all edges present ✓
Only ASR checks full ordered sequence

ASR Results — T3 Error Analysis

Table 4.15: Per-sample T3 hidden deviation analysis (single representative repeat, 250 samples per model).

A hidden deviation is a sample where TSR succeeds but ASR < 100%. Hidden rate = hidden deviations / successful samples.

Model	TSR	ASR	Failures	Hidden	Perfect	Hidden Rate
Zero hidden deviations						
Gemma4:e2b	100.0%	100.00%	0	0	250	0.0%
Gemma4:e4b	100.0%	100.00%	0	0	250	0.0%
Gemma4:26b	100.0%	100.00%	0	0	250	0.0%
Gemma4:31b	100.0%	100.00%	0	0	250	0.0%
GPT-OSS:20b	100.0%	100.00%	0	0	250	0.0%
GPT-OSS:120b	100.0%	100.00%	0	0	250	0.0%
MSmall3.2:24b	100.0%	100.00%	0	0	250	0.0%
Minimal hidden deviations (≤ 2)						
Qwen2.5:32b	100.0%	99.96%	0	1	249	0.4%
Qwen3:32b	100.0%	99.96%	0	1	249	0.4%
Qwen3:8b	100.0%	99.96%	0	1	249	0.4%
Qwen3:14b	99.6%	99.56%	1	1	248	0.4%
Llama3.1:70b	99.6%	99.38%	1	2	247	0.8%
Qwen2.5:14b	99.6%	99.51%	1	2	247	0.8%
Llama3.1:8b	94.8%	94.64%	13	2	235	0.8%
Significant hidden deviations						
Magistral:24b	94.4%	93.42%	14	22	214	9.3%
GPT-5.2	99.6%	97.20%	1	54	195	21.7%
GPT-4.1	100.0%	97.24%	0	62	188	24.8%
Qwen2.5:7b	52.4%	47.09%	119	118	13	90.1%

Three Severity Bands

Per-model hidden-rate groups:
Zero / Minimal / Significant
surfaces silent compliance
risk per model

Frontier Models at Risk

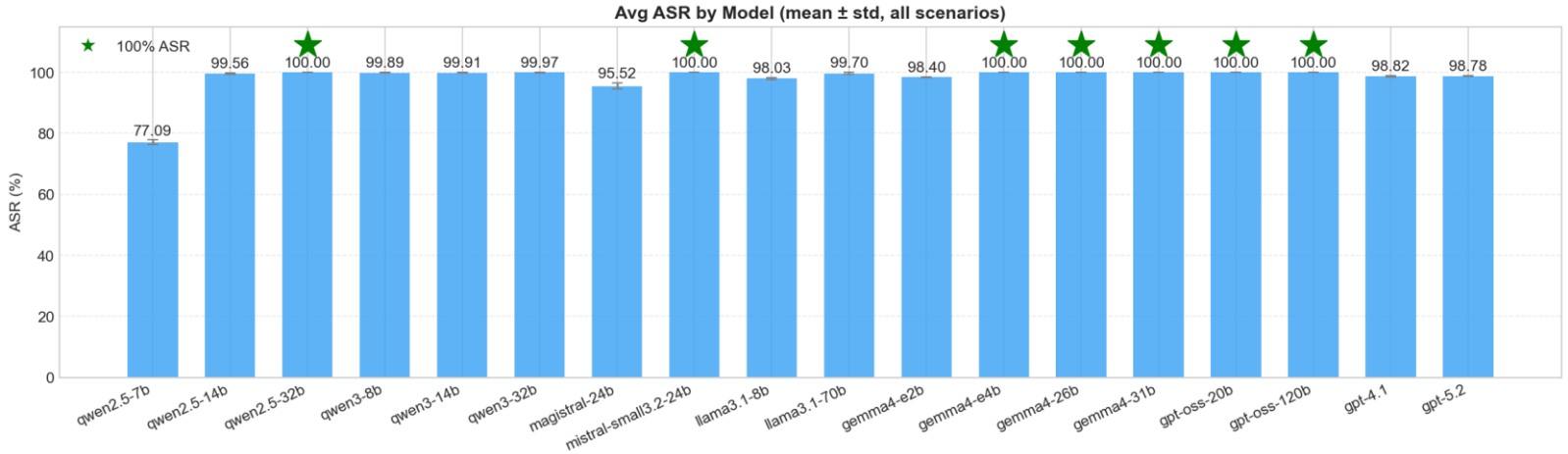
GPT-5.2 hidden rate = 21.7%
GPT-4.1 hidden rate = 24.8%

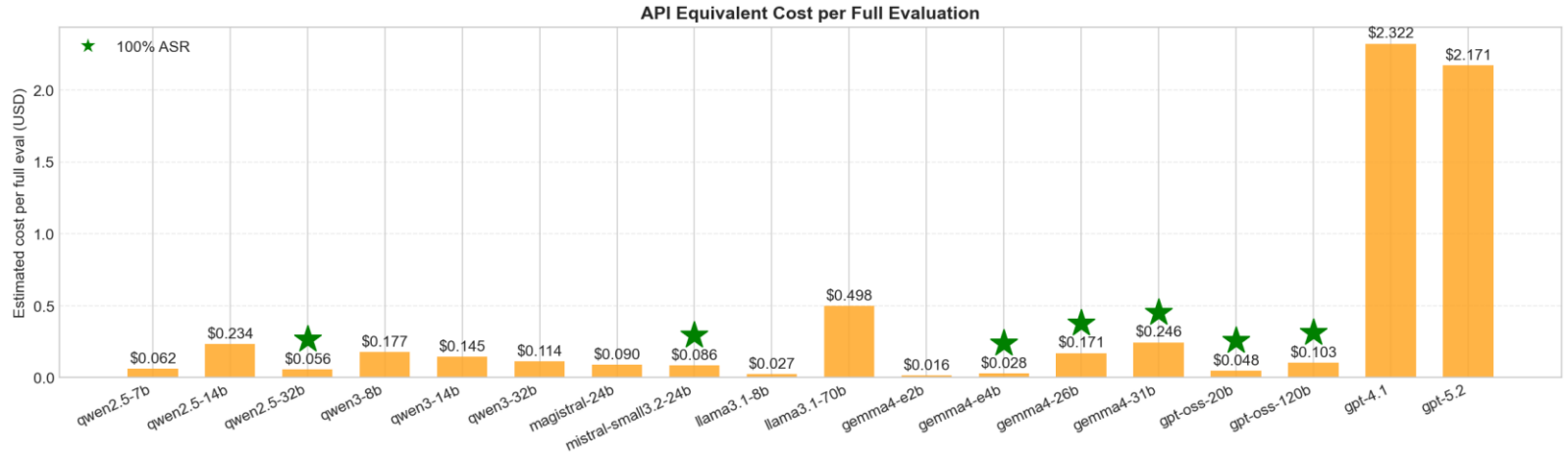
all pass TSR-only evals

Why It Matters

>1 in 10 "successes" silently
skip the CPA confirmation
TSR misses it —
ASR surfaces it

ASR Results — Cost / Benefit Analysis





Note: Ollama models run locally — actual cost is $0. OpenRouter prices shown for reference.

Key Findings

- **Best value:** gemma4-e4b achieves 100% ASR at only $0.007/run — 83x cheaper than GPT-4.1
- **GPT premium:** GPT-4.1 ($0.58/run) and GPT-5.2 ($0.54/run) cost 50–80x more than top open-weight models
- **Token volume:** Most models use 5.4–6.2M input tokens across 20 runs; Qwen3 and Gemma4-31b generate 2–3x more output tokens
- **Size ≠ quality:** llama3.1-70b costs $0.5/run but scores lower ASR than several sub-$0.05 models

HMASP Architecture V3 – Production-grade Agentic Payment Solution

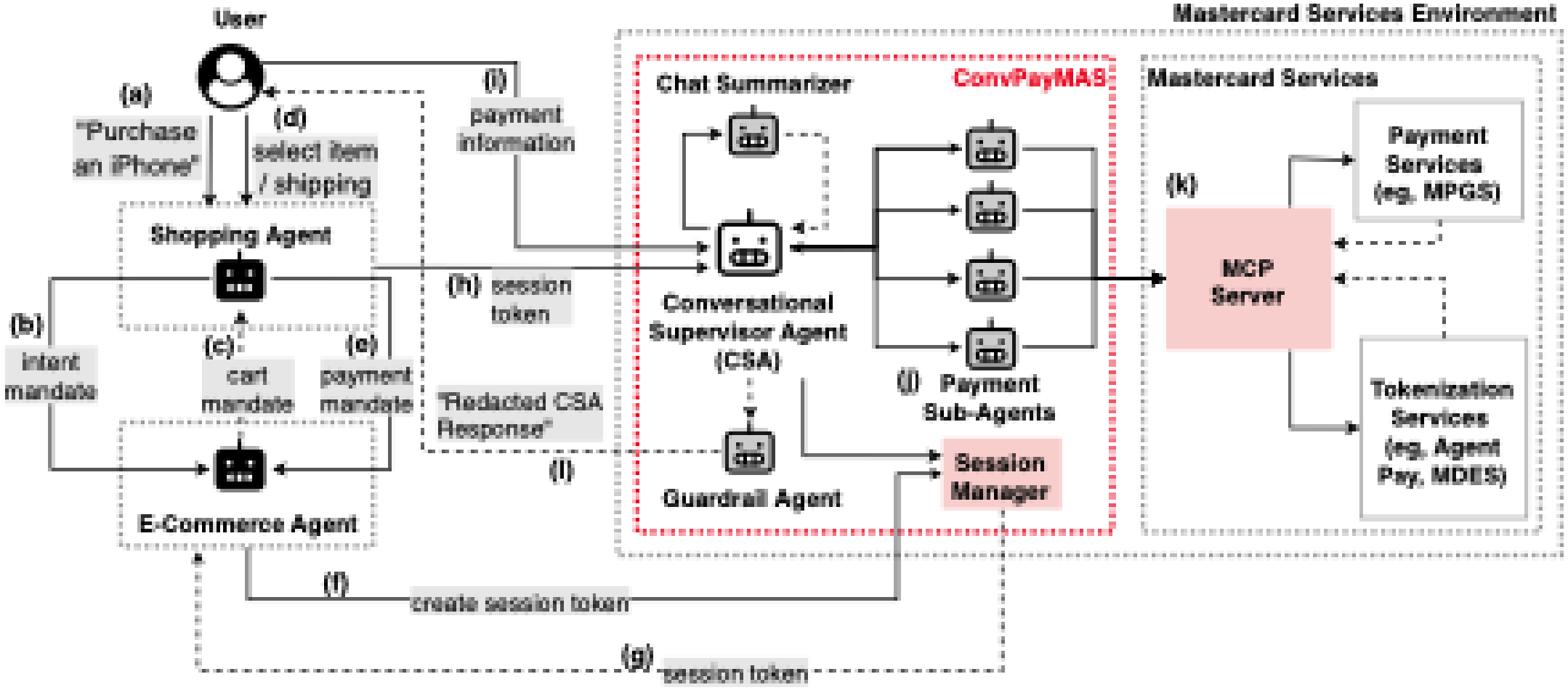


Figure 4.5: Overview of the agentic payment ecosystem with ConvPayMAS.

Accepted at AAMAS 2026 (25th International Conference on Autonomous Agents and Multi-Agent Systems) Demonstration Track

Enterprise Agentic Eval Framework

- Enterprise Agentic Evaluation & Optimization Framework (in development) — built on ConvPayMAS + ASR diagnostics:
  - Pillar 1 — Tool-Call Reliability (structured error taxonomy)
  - Pillar 2 — Workflow Fidelity (ASR · TSR · HF1)
  - Pillar 3 — Cost Tracking (token · API cost · latency)
- Closed loop: Evaluate → Diagnose → Optimize across the full agentic pipeline.

1 EVALUATE

Multi-Metric Assessment

Run 150K+ instances across 30 LLMs

Collect TSR, HF1, and ASR concurrently

Measure on consumer HW (M5 Max) and cloud (H100)

2 DIAGNOSE

Trajectory-Level Analysis

ASR decomposes into Step Recall × Precision

Pinpoint exact agent where workflow deviates

Map to structured error taxonomy (Paper 5)

3 OPTIMIZE

Targeted Interventions

Per-agent prompt refinement at failure point

Deterministic guards for unreliable LLM routing

Re-evaluate to verify improvement w/o regression (typically 2–3 cycles)

Pillar III — Enterprise Deployment & Applications

Chapter 5 · focus: Edge · DeepSeek-R1 · Task Complexity · Entity Matching

LLMs at the Edge with Ollama (IJCNN 2025)

- Problem: energy footprint & data-residency push enterprises to deploy LLMs locally — but on which hardware, with which model size, and with what trade-offs?
- Study design: 14 open-weight models (Qwen2.5 and Llama3 families, 0.5B → 90B params) × 6 hardware platforms, all via Ollama with Q4_K_M quantization.
- Hardware platforms benchmarked:
  - ➢ High-end desktop GPU — NVIDIA RTX A6000 Ada (48 GB, 18,176 CUDA)
  - ➢ Consumer desktop GPU — NVIDIA RTX 4090 (24 GB)
  - ➢ Laptop GPU — NVIDIA RTX 4090 Laptop (16 GB)
  - ➢ Apple Silicon — M3 Max (96 GB unified) · M4 Max (48 GB unified)
  - ➢ Edge device — NVIDIA Jetson AGX Orin Developer Kit (64 GB, up to 275 TOPS)
- Evaluation metrics: F1 score · Energy per Inference (J) · Performance-to-Power Ratio (%/J) · Throughput (tokens/s).

- Paper: P10 — LLMs at the Edge with Ollama, IJCNN 2025

Edge Deployment — Results & Guidelines

- 1) Quantization — Ollama Q4_K_M beats bitsandbytes 4-bit:
  - llama3.1:8b 92.87% F1 (+3.37 pp over baseline 89.50%)
  - llama3.1:70b 95.73% F1 (+0.22 pp over baseline 95.51%)
- 2) Platform matters as much as the model — WSL delivers up to 21× throughput over native Windows:
  - qwen2.5:0.5b WSL 16,933 t/s vs Windows 816 t/s (21×)
  - gap narrows to ~15% at 90B — smaller models benefit most from WSL.
- 3) Optimal model sizing — medium (7B–32B) wins on performance / efficiency:
  - qwen2.5:32b 96.75% F1 (±0.11) · ~710 t/s → accuracy leader in practical regime
  - llama3.1:8b 92.87% F1 (±0.46) · ~3,017 t/s · 7B–8B models hit the sweet spot
  - qwen2.5:3b 91.00% F1 · 558 J/inference · highest performance-to-power ratio (0.163 %/J)
- 4) Cross-platform stability — larger models are more consistent (σ 0.07–0.23 for 70B+ vs 0.26–0.90 for small).
- Deployment guidance — 7B–8B for latency-sensitive workloads; 32B for accuracy-sensitive; small (≤3B) for energy-constrained edge.

Explainable Sentiment Analysis with DeepSeek-R1 (IEEE Intelligent Systems)

- Motivation: LLM-based sentiment analysis needs to balance accuracy, efficiency, and explainability.
- DeepSeek-R1 is an open-source reasoning-native model trained with reinforcement learning — it generates step-by-step reasoning traces during inference, providing built-in explainability without external rationalization.
- Study: evaluate DeepSeek-R1 (full 671B and distilled 8B/14B/32B/70B variants) against GPT-4o and GPT-4o-mini on three sentiment tasks of increasing complexity:
  - IMDB — binary sentiment (simple)
  - Amazon Reviews — 5-class sentiment (moderate)
  - GoEmotions — 27-class emotion recognition (complex)
- Few-shot prompting sweep: 0 → 50 shots.

- Paper: P13 — DeepSeek-R1 Explainable Sentiment, IEEE Intelligent Systems ★ first first-author journal publication

DeepSeek-R1 — Findings (IEEE Intelligent Systems)

- Headline results:
  - Amazon 5-class: DeepSeek-R1 91.39% F1 @ 30 shots vs GPT-4o 87.99% @ 40 shots.
  - IMDB binary: DeepSeek-R1 99.31% accuracy @ only 5 shots · exceptional few-shot efficiency.
  - GoEmotions 27-class: 45.33% accuracy @ 10 shots · on par with GPT-4o 45.17%.
- Architecture over scale:
  - deepseek-r1:32b (81.7% F1) > deepseek-r1:70b (74.2% F1)
  - 32B Qwen2.5-based distill beats 70B Llama-based distill by 6.69 pp on Amazon Reviews.
  - Distillation quality depends on base architecture compatibility — not raw parameter count.
- Explainability — unique strengths of reasoning-native architecture:
  - Dual-layer output: concise classification + full step-by-step reasoning trace.
  - Traces temporal progression in mixed reviews (negative experience → resolution).
  - Greater sensitivity to emotional language and indirect reporting — nuanced sentiment detection.
- Throughput trade-off: reasoning reduces throughput (334 t/s vs GPT-4o's 1,220 t/s) — acceptable for auditability-first use cases.

Task Complexity Matters (PAKDD 2026)

- Central question: Do reasoning capabilities justify their computational overhead across sentiment tasks of varying complexity?
- First large-scale, cross-architectural comparison of reasoning vs non-reasoning modes — 504 unique configurations:
  - 7 model families — DeepSeek-R1 · Qwen3 · Granite3.3 · Magistral · DeepSeek-V3 · LLaMA3 · Qwen2.5
  - 3 datasets — IMDB (binary) · Amazon Reviews (5-class) · GoEmotions (27-class)
  - 7 shot levels — 0 · 5 · 10 · 20 · 30 · 40 · 50
  - Thinking / Non-thinking variants for Qwen3, Granite3.3, Magistral.
- Task-complexity hypothesis — validated empirically:
  - IMDB (binary): mean F1 gap Distilled − Base = −4.8 pp · 100% failure rate · reasoning HURTS.
  - Amazon (5-class): mean F1 gap = −3.6 pp · 80% failure rate · reasoning mostly hurts.
  - GoEmotions (27-class): mean F1 gap = +2.0 pp · 14% failure rate · reasoning HELPS substantially.
- Extreme cases — reasoning degrades binary sentiment by up to −19.9 F1 pp; reasoning improves 27-class emotion by up to +16.0 F1 pp.
- Speed overhead of thinking mode: 1.0× (Granite) → 11.5× (Qwen3-32B) → 54.4× (Magistral-24B on GoEmotions).
- Deployment guidance — use base/non-thinking models for simple tasks; reserve reasoning for complex multi-class problems where 2–4 pp gains justify 2.1–54× compute overhead.

- Paper: P14 — Task Complexity and Reasoning Effectiveness, PAKDD 2026

Entity Matching with LLMs at Scale (PAKDD Workshop 2026)

- Problem: merchant-data enrichment at Mastercard (GAIME) requires matching billions of noisy merchant strings to canonical entities — classical methods plateau.
- Study design — one of the largest cross-architectural LLM entity-matching evaluations to date:
  - 46 model configurations × 8 benchmark datasets (Abt–Buy, Amazon–Google, WDC, ...) using the ComEM framework.
  - Spanning GPT-5/5.2, GPT-OSS (open-weight), Claude, Kimi, GLM, Qwen3, DeepSeek, Gemma, Llama4, Granite4, MiniMax — proprietary and open-weight.
  - Configurable reasoning effort (low/medium/high) and thinking toggles (on/off/adaptive) across families.
- Key findings:
  - Open-weight models rival proprietary APIs — GPT-OSS:120b(high) achieves 92.10 F1 at ~20× less cost than GPT-5(high).
  - Reasoning effort has opposite effects across families — low is optimal for GPT-5; high is critical for GPT-OSS; Claude shows the most dramatic impact (+10.13 F1 for Sonnet 4.5 with thinking, yet −0.25 for Opus 4.5); Qwen3-235b drops 5.09 F1 with thinking.
  - Pareto frontier dominated by GPT ecosystem — ultra-budget models under $0.10 reach up to 89 F1.
  - Backward link to P14 — confirms model-dependent reasoning effects on a non-sentiment, real-enterprise task.
  - Forward link — directly informs production model selection inside GAIME at Mastercard.

- Paper: P15 — Entity Matching with LLMs at Scale: Accuracy, Cost, and Reasoning Effects, PAKDD Workshop 2026

Response to Examiner Comments

Four recommended minor changes → ten targeted revisions

Responses to Examiner Comments (1 / 2)

- Comment 1 — Highlight the most focused work with more content.
  - Added Research Journey narrative in Chapter 1 with a Research Progression figure covering all 17 papers and three cross-pillar arrows.
  - Replaced Section 4.8 (EdgeAgent) with a comprehensive ASR section: 18 LLMs × 90,000 instances, formal metric, regulatory discussion, per-sample deviation analysis.
  - Added Section 5.4.3 — Entity Matching with LLMs at Scale: 46 configurations × 8 datasets, Pareto frontier analysis, family-dependent reasoning effects.
  - Added Ongoing Work subsection — ConvPayMAS + AP2 three-mandate verification + enterprise agentic evaluation framework.
- Comment 2 — Theoretical contributions should be clearer.
  - ASR now presented with full formal definition grounded in process-mining conformance checking (van der Aalst, Carmona et al.) — bridges process mining and LLM evaluation.
  - Structured error taxonomy reframed as a generalisable theoretical framework: four error types × N tool stages scales with system complexity.
  - Entity matching family-dependent reasoning effects elevated as a theoretical finding beyond the specific domain.

Responses to Examiner Comments (2 / 2)

- Comment 3 — Claims about generalizability should be more cautious.
  - "12-category error taxonomy" renamed to "structured error taxonomy".
  - Explicit note added — the 12 categories arise from 4 error types × 3 tool-call stages in the invoice reconciliation system; other systems would yield different counts.
- Comment 4 — Stronger linkage and integration across chapters.
  - New Research Progression figure with explicit cross-pillar arrows: P3 → P4 (quality metrics) · P10 → P4 (edge deployment) · P14 → P7 (reasoning effectiveness).
  - Entity Matching section (Ch 5) designed with explicit backward link to Section 5.3.3 (task-complexity findings) and forward link to GAIME production system.
  - ConvPayMAS (Ch 4) serves as an integration point: ASR diagnostics → production payment system → enterprise evaluation framework — bridges Chapter 4 (evaluation) to Chapter 5 (deployment).
  - Cross-file consistency review (Change 8) aligned publication counts, metric definitions, and model results across Chapters 1, 4, 5, 6, and the abstract.
- Revised dissertation contains 17 peer-reviewed publications (11 published · 6 accepted) with substantially deeper analysis on the most focused contributions.

Conclusion

- Thesis statement: Open-source LLMs, when optimized and evaluated with reliability-aware metrics, can support privacy-preserving and cost-effective enterprise deployment.
- Delivered across three interconnected pillars:
  - Pillar I — RAG Optimization: RAP metric (NAACL '25) + ReDA tuning → production deployment in MAP at Mastercard.
  - Pillar II — Agentic AI: HMASP (PAKDD '26) + ASR (PAKDD Wksp '26) + ConvPayMAS (AAMAS '26 demo) → Agent Pay.
  - Pillar III — Deployment & Applications: Edge LLMs (IJCNN '25) + DeepSeek-R1 (IEEE Intelligent Systems) + Task Complexity (PAKDD '26).
- 17 peer-reviewed publications · 7 novel metrics · 4 architectural patterns · 3 production deployments at Mastercard · 1 patent-pending system.
- Core insight: architectural design and reliability-aware evaluation dominate raw model capacity — open-weight models repeatedly match or exceed proprietary APIs when tuned and measured correctly.
- The contributions provide an empirically grounded foundation for building robust, auditable generative-AI systems in real enterprise settings.

Industry Impact at Mastercard

- MAP — Mastercard Assistant Platform
  - RAP metric and evaluation-guided tuning procedure deployed as the primary output-quality gate for Mastercard's internal RAG knowledge assistant.
  - First dissertation contribution to transition from NAACL '25 research into production at scale.
- Conversational Payment Agent — Mastercard’s future-proof agentic commerce offering
  - ASR provides the trajectory-level visibility that regulated industries require for agentic payment flows.
  - ASR-guided engineering yields +24.2 pp average TSR improvement across 11 production models in HMASP.
  - ConvPayMAS (demonstrated at AAMAS 2026) is the production evolution — AP2 three-mandate verification + patent-pending security features.
- GAIME — Generative AI Merchant Enrichment
  - Entity-matching accuracy–cost–reasoning trade-offs (P15) directly inform model selection for merchant-data enrichment.
  - Ongoing LoRA / QLoRA fine-tuning of sub-4B-parameter models is extending these findings toward low-cost, latency-sensitive production serving.
- Broader impact — this research reduces three barriers to enterprise LLM adoption:
  - open-weight models viable for production
  - edge deployment practical on consumer GPUs
  - reliability-aware evaluation covers what accuracy alone misses

Limitations

- Language coverage — enterprise datasets used are predominantly English; multilingual enterprise generalization needs further study.
- Simulation vs production — agentic evaluations (HMASP, CMAS4G2, invoice reconciliation) rely on simulated workflows; production validation is ongoing via ConvPayMAS.
- Task scope of the complexity hypothesis — validated on classification; extension to open-ended generation tasks remains future work.
- ASR assumes a specifiable expected trajectory — fully open-ended autonomous agents may need extensions to the metric.
- Cost analyses reflect public API pricing as of Q1 2026 — absolute numbers will shift over time, but the Pareto structure should remain informative.

Future Work

- ConvPayMAS — productize the AP2 three-mandate verification architecture with patent-pending cryptographic security; comprehensive evaluation, scalability testing, and regulatory certification.
- Enterprise Agentic Evaluation & Optimization Framework — generalize ASR, ECR@1, and the structured error taxonomy into a unified three-pillar toolkit (tool-call reliability · workflow fidelity · cost tracking).
- LoRA / QLoRA fine-tuning of sub-4B-parameter models for low-latency, low-cost production serving (extending GAIME work).
- Extend ASR to open-ended agentic tasks where expected trajectories are partial or learned rather than fully specified.
- Continuous benchmarking as new reasoning-model families emerge — keep task-complexity guidance current.

Publications Supporting the Dissertation (1 / 2)

- 17 peer-reviewed publications · 11 published · 6 accepted

- Pillar I — RAG Optimization (Chapter 3)
  - ➢P1 — Orca-2 in RAG Applications · PAKDD Workshop 2024
  - ➢P2 — Llama-2 in Retrieval-Augmented Generation · IEEE CAI 2024
  - ➢P3 — Repetition-Aware Performance (RAP) Metric · NAACL 2025

- Pillar II — Agentic AI (Chapter 4)
  - ➢P4 — Multi-Agent Invoice Reconciliation · ICDSAAI 2025
  - ➢P5 — Diagnostic Framework for Tool Invocation Reliability · ICAIBD 2026
  - ➢P6 — LLM-as-a-Judge with ECR@1 · AAAI Workshop 2026
  - ➢P7 — CMAS4G2 — Cooperative Gherkin Generation · PAKDD Workshop 2026
  - ➢P8 — HMASP — Hierarchical MAS for Payments · PAKDD 2026
  - ➢P9 — ASR — Agentic Success Rate · PAKDD Workshop 2026

Publications Supporting the Dissertation (2 / 2)

- Pillar III — Enterprise Deployment & Applications (Chapter 5)
  - P10 — LLMs at the Edge with Ollama · IJCNN 2025
  - P11 — Reasoning Sentiment Analysis (ICDM Workshop) · ICDM Workshop 2025
  - P12 — Logical Reasoning via Lateral Thinking Puzzles · IEEE SSCI 2025
  - P13 — Explainable Sentiment Analysis with DeepSeek-R1 · IEEE Intelligent Systems
  - P14 — Task Complexity and Reasoning Effectiveness · PAKDD 2026
  - P15 — Entity Matching with LLMs at Scale · PAKDD Workshop 2026
  - P16 — Chinese–English Translation Optimization · IEEE SSCI 2025
  - P17 — LLM Sentiment for Sports Predictive Analytics · ICDM Workshop 2025

- Summary statistics
  - 17 papers (first author): 5 Rank A (P3, P8, P10, P13, P14), 2 Rank B (P12,P16)
  - 7 novel metrics
  - 4 architectural patterns
  - 3 Mastercard production deployments
  - 1 patent-pending system (ConvPayMAS)

Acknowledgments

- Dissertation Committee
  - Assoc. Prof. Zhaoxia WANG — Supervisor / Chair, SMU SCIS
  - Prof. Chong Wah NGO — Co-Supervisor / Co-Chair, SMU SCIS
  - Assoc. Prof. Qianru SUN — SMU SCIS
  - Prof. Chunyan MIAO — External Reviewer, NTU
- Funding & Institutional Support
  - School of Computing and Information Systems, Singapore Management University.
  - Singapore Digital Scholarship (Postgraduate), IMDA, 2025.
  - Mastercard — R&D collaboration, computational resources, and conference support.
- Co-authors across 17 publications — thank you for your contributions and collaboration.
- Family and friends — for their unwavering support throughout this EngD journey.

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Dissertation Publications (10 Focus Papers)

- [P1] Huang, D. & Wang, Z. (2024). Evaluation of Orca 2 Against Other LLMs for Retrieval Augmented Generation. PAKDD Workshop 2024.
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